

Mobile Application Development



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Mobile Thinking & Application Engineering



A mobile application isn't just a collection of screens.

It is software running inside a constrained, connected, sensor-rich physical device.

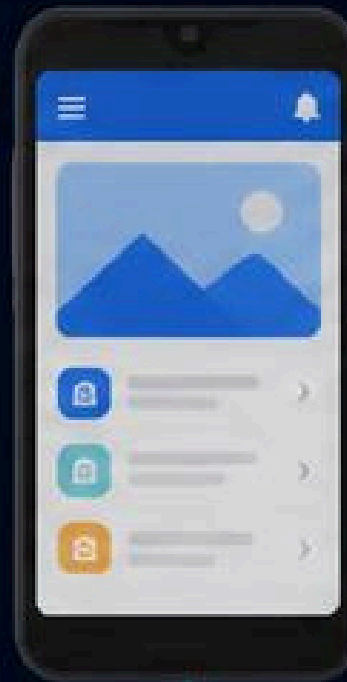
MOBILE DEVICES COME WITH UNIQUE CONSTRAINTS

-  **Screen Size & Density**
Vast range of sizes and pixel densities
-  **Touch Targets & Gestures**
Precision matters on small screens
-  **Battery & Performance**
Limited power and compute resources
-  **Network Variability**
From 2G to 5G, connectivity is unpredictable
-  **Permissions & Privacy**
Access to data and hardware is restricted
-  **Sensors & Hardware**
Accelerometer, GPS, Camera, etc.



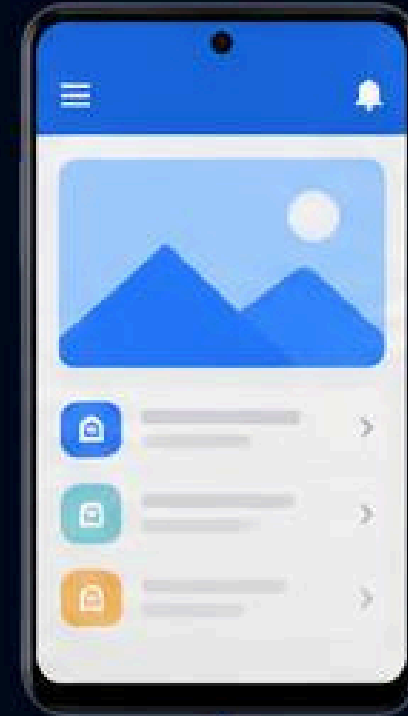
Budget

Basic & Essential



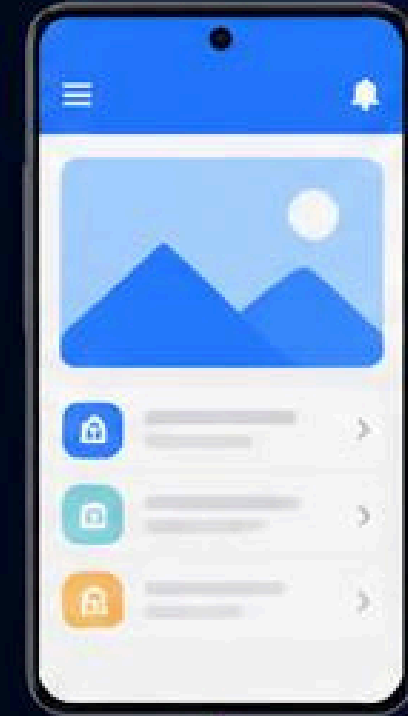
Mid-range

Balanced & Capable



Flagship

High-end & Powerful



Same Application



Different Capabilities



Great mobile developers design and build with constraints in mind, so the experience is reliable, inclusive, efficient, and delightful—on every device.



Responsive UI



Adaptive Performance



Resilient Connectivity



Secure & Respectful



Inclusive Experience



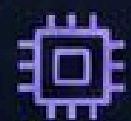
Performance-aware development

Performance is a **requirement**, not a finishing touch.



Every decision has a consequence.

UI choices, asset sizes, state management, rebuilds, network calls—everything impacts performance.



Design with the whole spectrum in mind.

Your app will run on flagship phones and budget devices alike.



Great apps don't just perform well on ideal devices.

They adapt, remain usable, and earn trust.



Performance isn't about doing less.

It's about **doing what matters, everywhere.**



Faster start
Better first impression



Lower usage
Respect battery & data



Happier users
Smoother experiences build loyalty



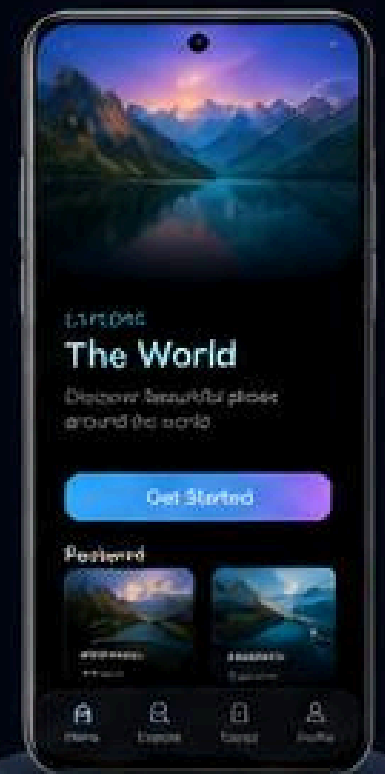
Better results
Performance drives retention

One app. Many devices.

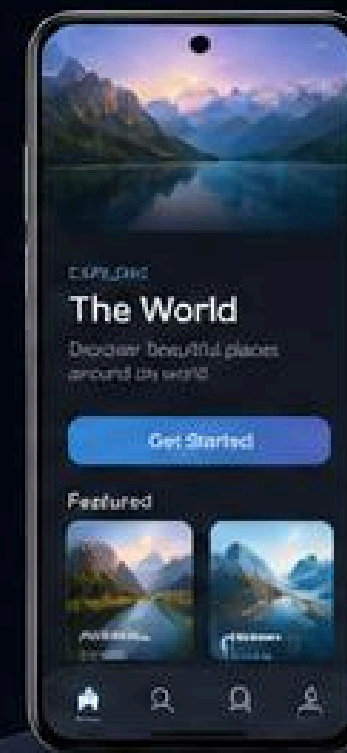
Degrades gracefully.

High-end
(More capable)

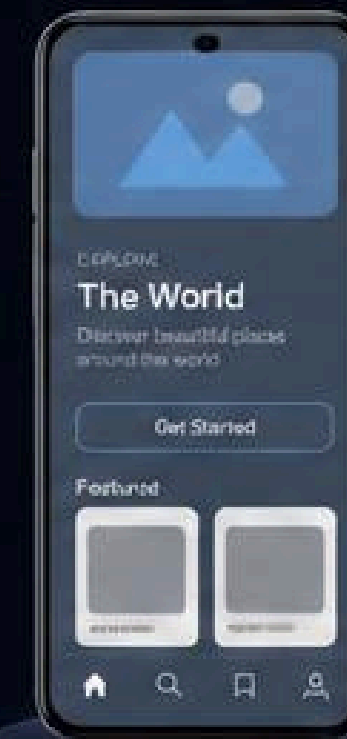
Low-end
(Less capable)



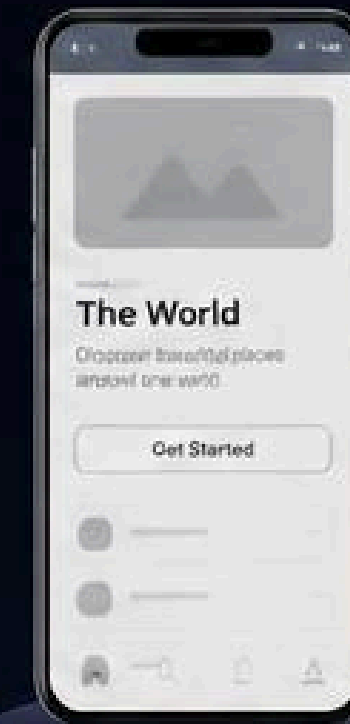
Rich visuals
Smooth animations
All features



Optimized visuals
Light animations
Core features



Minimal visuals
Essential animations
Essential features



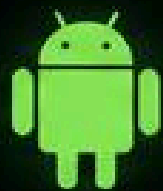
Lightweight UI
No extra effects
Always functional

Performance-Aware Development

CLIENT REQUEST

Mobile Application

- ✓ Blazing Fast
- ✓ Instant Loading
- ✓ For Android Platform



Which approach
should I choose?



Flutter Development



Blazing Fast?



Instant Loading?



Cross-Platform Advantage



OR



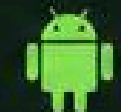
Native Android Development



Blazing Fast



Instant Loading



Built for Android



Choose the right approach.

Based on requirements. Driven by performance.

When performance is critical,
Native wins.



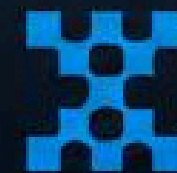
Do I Need to Know All These Chipsets?



So Many Chipsets. So Many Details.



Snapdragon
8 Gen 3



Exynos
2400

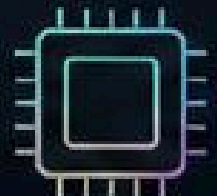
5G



MediaTek
Dimensity 9300



A16
Bionic



And Many
More...

Different architectures. Different capabilities. **Do I need to master all of them?**



No.

Flutter and the **Operating System** provide abstraction layers that shield app developers from most low-level hardware details.



Focus on building **amazing apps**.
We handle the complexity underneath.

★ Write Once. Run **Anywhere**.

Offline-First Thinking

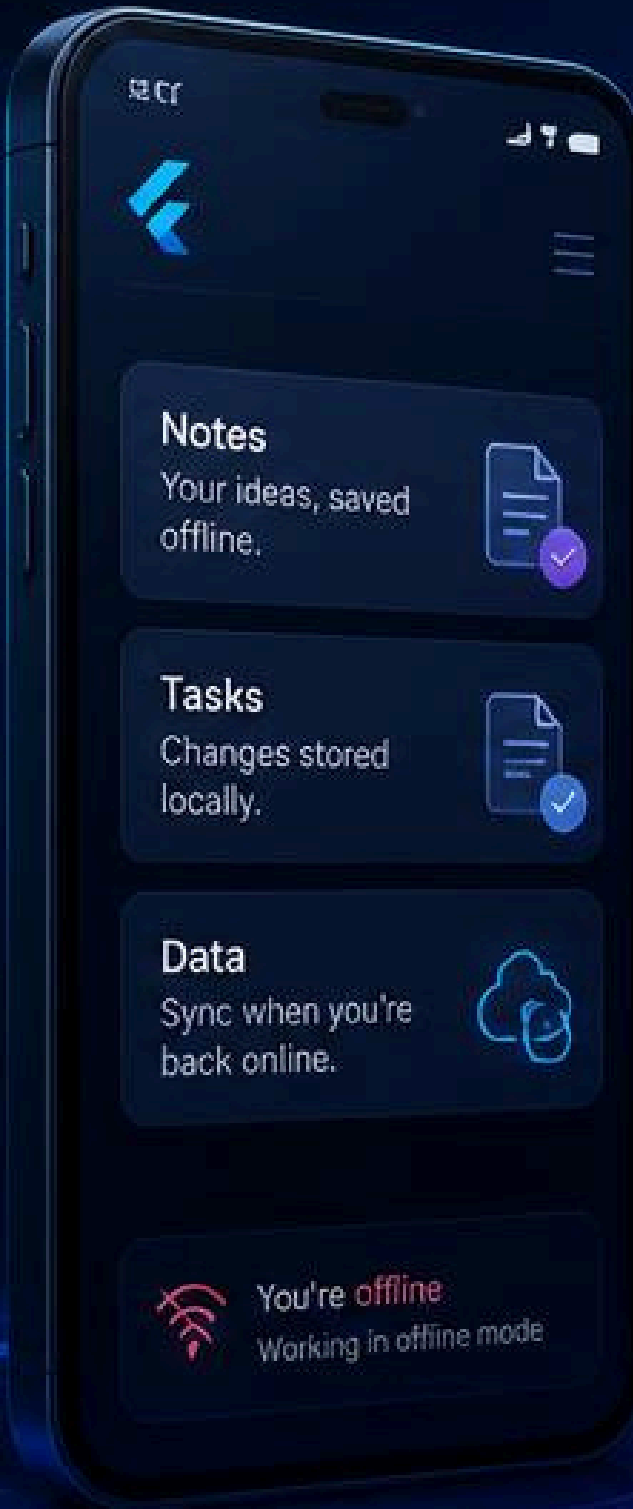


What if the internet Disappears?

Mobile devices are highly mobile. They rely on networks like 3G, 4G, or 5G — but coverage isn't guaranteed everywhere. Some places have weak signals. Some have none at all.



Should the application **stop working?**



Strong Signal
Connected



Weak Signal
Unreliable



No Signal
Disconnected



Great mobile apps keep working,
no matter the connection.



Design for offline.
Delight always.



Build apps that are
Resilient. Reliable. Ready.

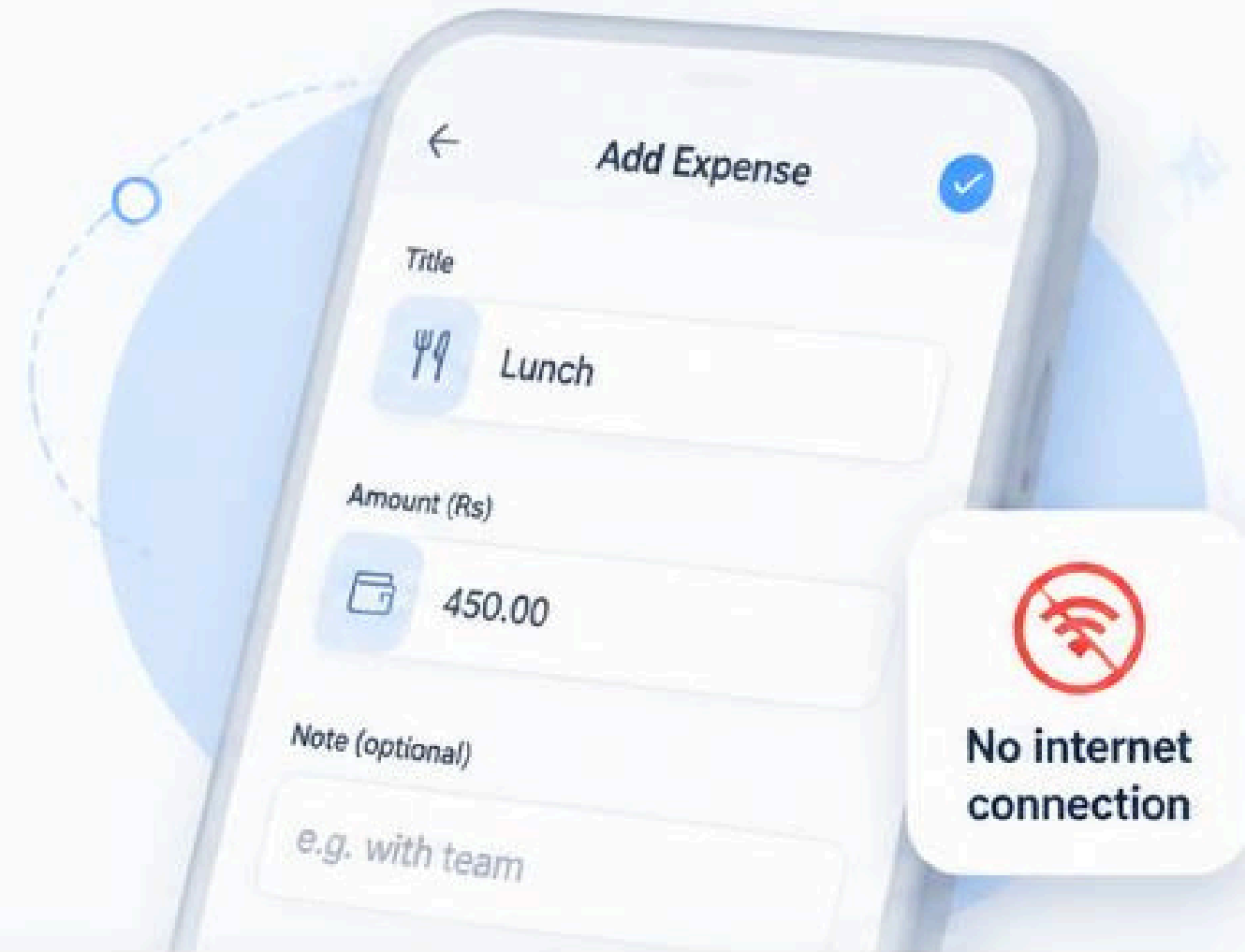


Imagine we're building an expense-tracking application



The user enters:

Lunch -> **Rs: 450.00**



✖ Bad Design



User



App



Enter data

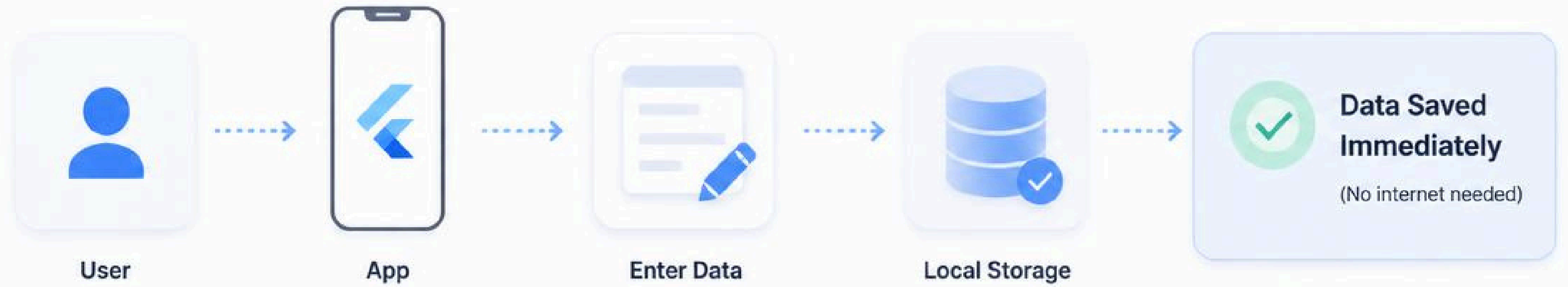


No Internet



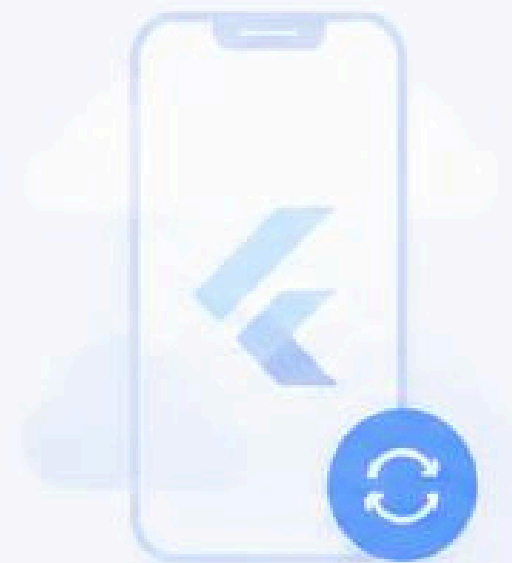
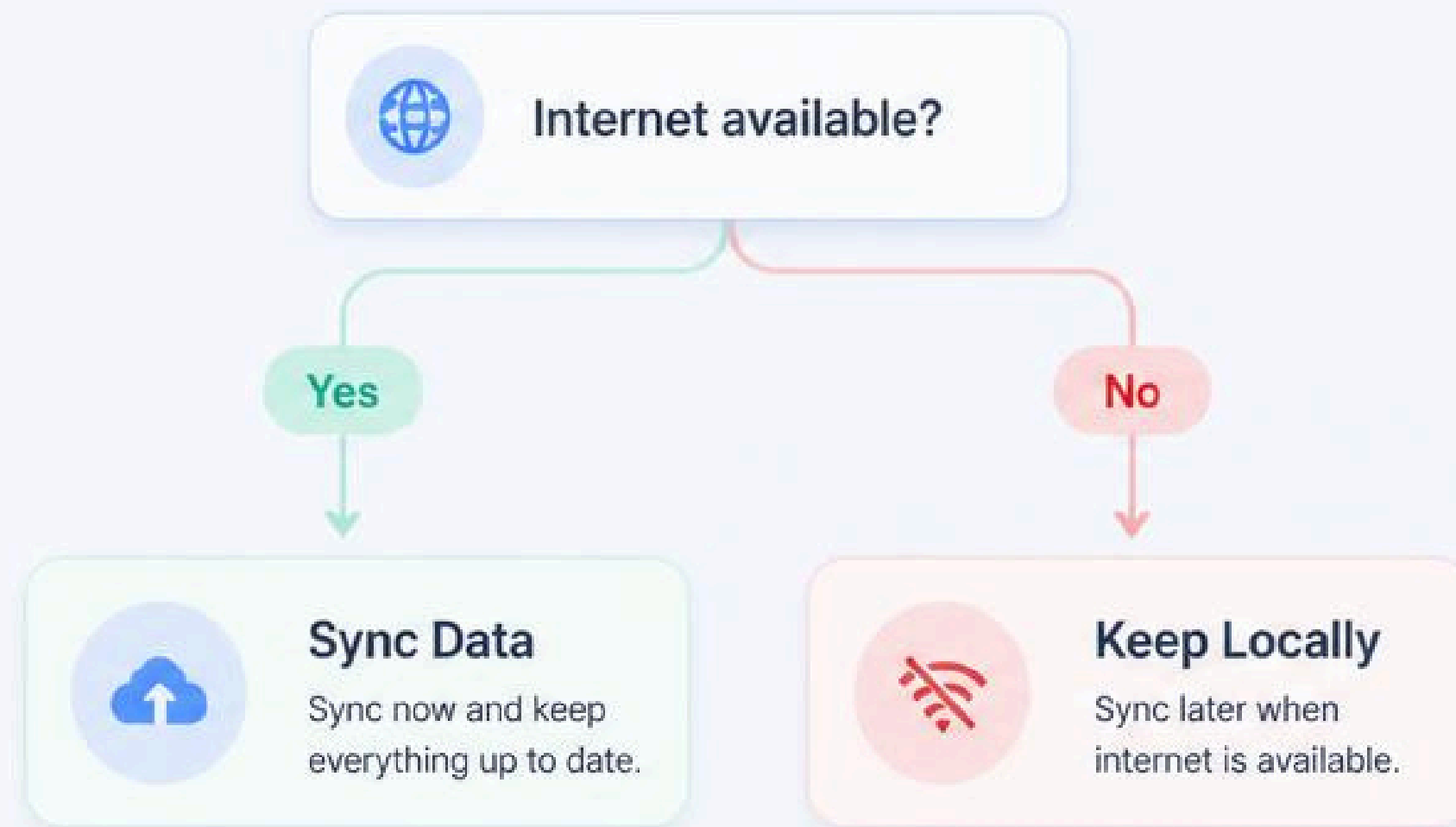
Data saving failed
(or waiting for
internet connection)





Cloud Synchronization

We design sync intelligently by checking the internet connection.



Offline-first is not just a feature, it's a mindset. Build apps that work for your users, **everywhere**.

Where Should Our Data Live?

To build truly reliable apps, we need to think about where our data should reside. Data can live **locally** on the device or **remotely** in the cloud.



Offline-first apps win 🏆

By keeping data close to the user, we ensure speed, reliability, and an exceptional user experience.

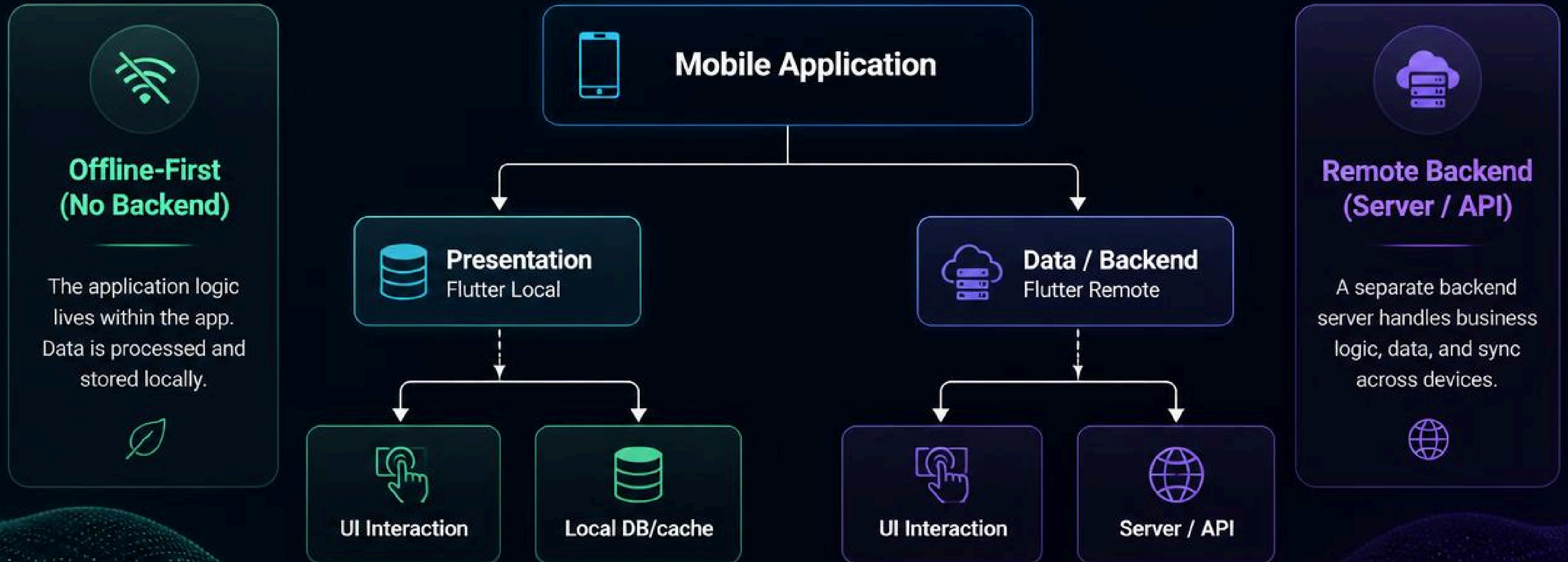


Choose the right storage for the right job. Combine local & remote to build powerful, resilient apps.



Networking & API Connectivity

Different apps, different backends. The architecture defines the experience.



★ Choose the right architecture for the right problem. **Offline. Online. Or both.**

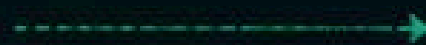
Backend Architecture: Local vs Remote



In mobile application development, the local-vs-remote decision is a **real design choice** with **real trade-offs** (offline-support, latency, data privacy, sync complexity), and picking wrong early leads to painful rewrites.

LOCAL-FIRST ARCHITECTURE

- ✓ Data stored and managed on the device



Local Database
(SQLite, Hive, etc.)

Great for offline support, faster access, and data privacy.

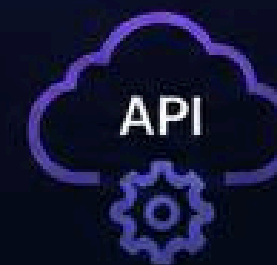
Vs

REMOTE-FIRST ARCHITECTURE

- ✓ Data stored and managed on a remote server



HTTP / HTTPS



Database
(MySQL, PostgreSQL, etc.)

Great for real-time data, scalability, and multi-device sync.



Remember: The right choice depends on your app's needs. Many mature apps use a **hybrid approach!**

Thank You!



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